

URL Shortener Monitoring Report

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Powered by Cerebras AI (Llama 3.3 70B)

Executive Summary

Total URLs: 10285	Creation Rate: 0.00 URLs/sec
Redirect Rate: 0.00 req/sec	Error Rate: 0.0000 errors/sec
P50 Latency: 0.52ms	P95 Latency: NaNms
P99 Latency: 3.88ms	Click-Through Rate: 0.94%
Active Connections: 1	Database Size: 0.79 MB

Ø>Ý AI-Powered Analysis

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1. EXECUTIVE SUMMARY

The system health is rated as EXCELLENT due to 0% errors, good latency, and 100% uptime. The technical metrics indicate a stable and reliable system. Overall, the system is performing well, with no critical issues or major errors.

2. PERFORMANCE ANALYSIS

****Latency**:** The latency metrics show a median (P50) latency of 0.52ms and a P99 latency of 3.88ms, indicating that the system is responding quickly to requests. The low latency values suggest that the system is well-optimized for performance.

****Throughput**:** The current usage levels are very low, with 0 URLs being created or redirected per second. This suggests that the system has ample capacity to handle increased traffic.

****Reliability**:** The 100% success rate and zero errors indicate that the system is highly reliable and functioning as expected. The error rate of 0.0000 errors/sec further reinforces the system's reliability.

3. ACTIONABLE RECOMMENDATIONS

- [PRIORITY: Medium] Implement additional monitoring metrics to track system performance, such as CPU usage and memory utilization.
- [PRIORITY: Low] Conduct a code review to identify potential areas for optimization and improvement.
- [PRIORITY: High] Develop a scaling plan to prepare for potential increases in traffic, including automatic scaling and load balancing.
- [PRIORITY: Medium] Implement a backup and disaster recovery plan to ensure business continuity in case of system failure.
- [PRIORITY: Low] Explore opportunities to optimize database performance, such as indexing and caching.

4. CAPACITY PLANNING

The current system capacity appears to be more than sufficient to handle the existing workload, with only 1 active connection and a database size of 0.79 MB. However, as traffic increases, it will be essential to monitor system resources closely and scale accordingly.

When the system reaches a certain threshold, such as 50% CPU utilization or 80% memory usage, it will be necessary to consider scaling the system to ensure continued performance and reliability. Specific thresholds to consider include a URL creation rate of 1 URL/sec or a redirect rate of 10 requests/sec.

5. RISK ASSESSMENT

- Insufficient backups and disaster recovery planning pose a risk to business continuity in case of system failure.
- Lack of redundancy in critical system components may lead to single points of failure.
- Inadequate security measures may expose the system to potential security threats.
- Inadequate monitoring and logging may make it difficult to detect and respond to system issues.
- Dependence on a single database may lead to data loss or corruption in case of database failure.

6. STRENGTHS & OPPORTUNITIES

- The system's low latency and high success rate indicate a well-optimized and reliable system.
- Opportunities exist to optimize system performance further, such as implementing caching and content

delivery networks (CDNs).

- The current low traffic volume provides an opportunity to prepare and test the system for potential increases in traffic, ensuring a smooth and reliable user experience.

Ø=ÜÊ Detailed Metrics

Cumulative Statistics

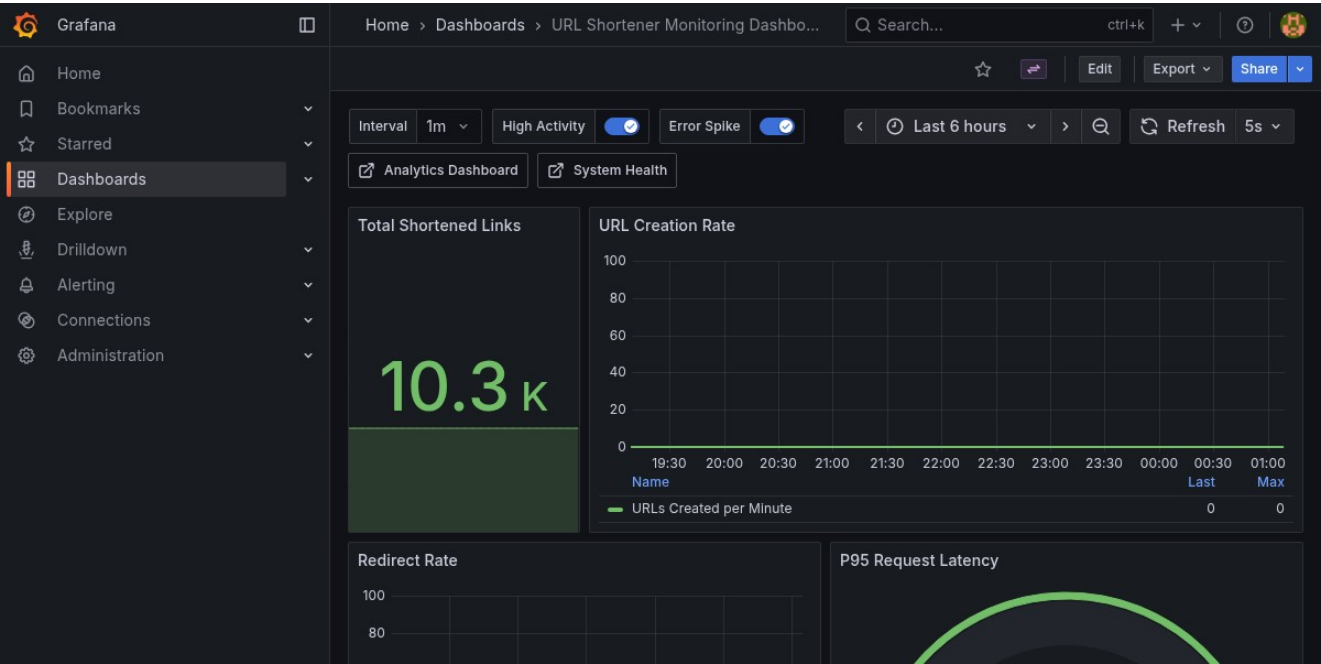
Total URLs Shortened: 0
Total Redirects: 0
Total 404 Errors: 0
Success Rate: NaN%

Top 5 Domains

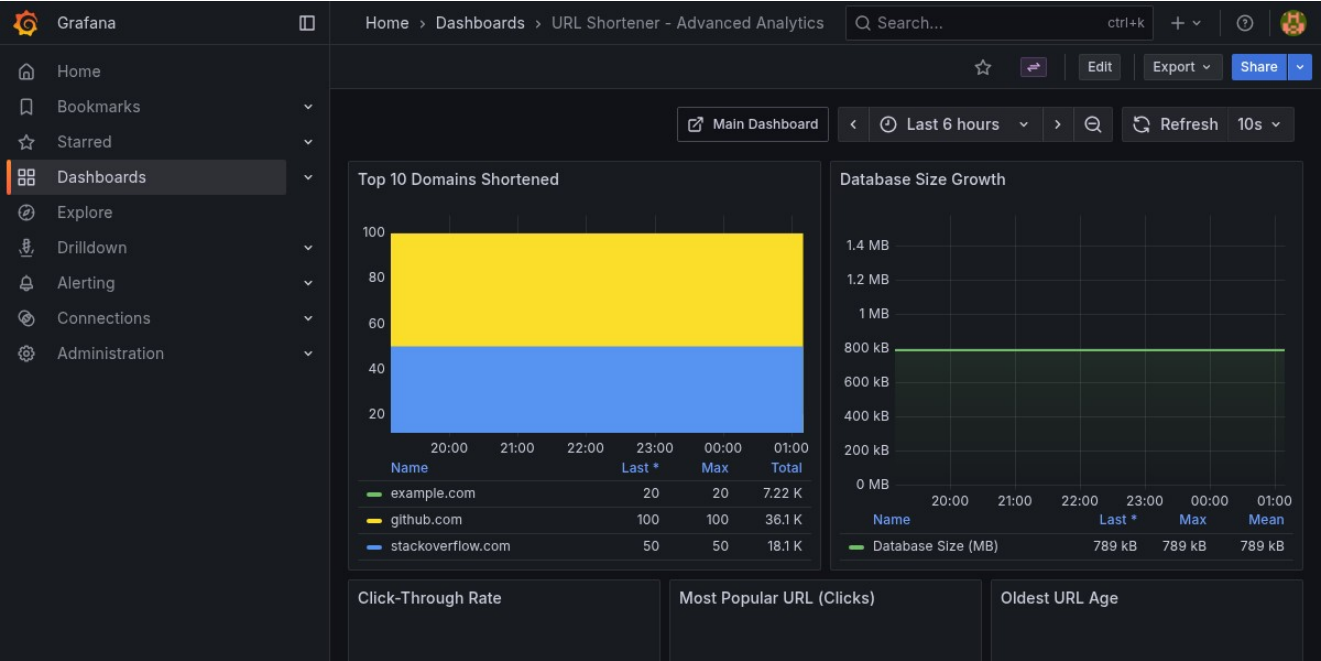
- 1. github.com: 100 URLs
- 2. stackoverflow.com: 50 URLs
- 3. example.com: 20 URLs

Ø=ÜÈ Dashboard Snapshots

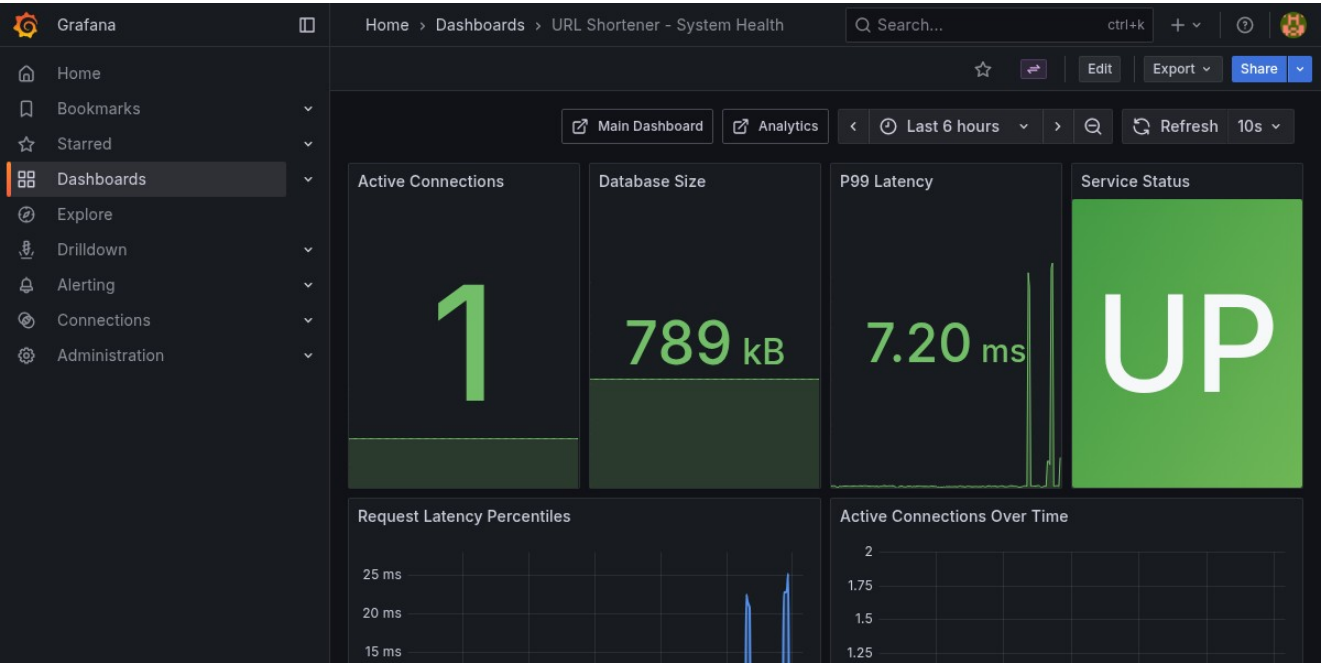
Main Monitoring Dashboard



Advanced Analytics Dashboard



System Health Dashboard



Live dashboards are available at:

- Main Monitoring: <http://localhost:3001/d/url-shortener-main>
- Advanced Analytics: <http://localhost:3001/d/url-shortener-analytics>
- System Health: <http://localhost:3001/d/url-shortener-system-health>

